

A SHARP 3-DEGREE ERDŐS–KO–RADO THEOREM FOR 4-UNIFORM FAMILIES

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ABSTRACT. For an intersecting family $\mathcal{F} \subseteq \binom{[n]}{4}$, we prove that its minimum triple degree satisfies $\delta_3(\mathcal{F}) \leq 1$ for every $n \geq 9$. The range is sharp: an explicit intersecting family on eight vertices has minimum triple degree two. The proof is elementary and self-contained. It treats $n = 9$ by a five-point link argument, while two disjoint triples outside a fixed edge settle every $n \geq 10$. As an independent check of the two values $n = 9, 10$ not covered by Huang and Zhang’s theorem, we also provide no-auxiliary CNF encodings and externally verified LRAT certificates for $n = 9, 10$.

1. INTRODUCTION

Write $[n] = \{1, \dots, n\}$. A family $\mathcal{F} \subseteq \binom{[n]}{k}$ is intersecting if every two of its members meet. For $S \in \binom{[n]}{d}$, set

$$d_{\mathcal{F}}(S) = |\{F \in \mathcal{F} : S \subseteq F\}|, \quad \delta_d(\mathcal{F}) = \min_{S \in \binom{[n]}{d}} d_{\mathcal{F}}(S).$$

The classical Erdős–Ko–Rado theorem is the case $d = 0$ of this degree viewpoint [1].

Huang and Zhang conjectured that, for $k > d \geq 0$ and $n \geq 2k + 1$, every intersecting $\mathcal{F} \subseteq \binom{[n]}{k}$ satisfies

$$(1.1) \quad \delta_d(\mathcal{F}) \leq \binom{n-d-1}{k-d-1}.$$

They proved (1.1) for $d \geq 2$ under the stronger hypothesis $n \geq 2k + 2d - 3$ [3]. At $(k, d) = (4, 3)$ their theorem begins at $n = 11$, whereas the conjectured range begins at $n = 9$. Theorem 1.2 of Gan, Han, and Im, specialized to $d = k - 1$, assumes $n \geq 14$ when $k = 4$, so it does not cover the endpoints $n = 9, 10$ [2].

We give a direct proof for the whole range $n \geq 9$, without using the large- n theorem. To the best of our knowledge, no earlier public source establishes either endpoint $n = 9$ or $n = 10$; our recorded search was completed on August 29, 2026.

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Theorem 1.1. *For every integer $n \geq 9$ and every intersecting family $\mathcal{F} \subseteq \binom{[n]}{4}$,*

$$\delta_3(\mathcal{F}) \leq 1.$$

Both the constant and the range are sharp.

The constant is attained by every full star. Section 4 records an exact $n = 8$ family with $\delta_3 = 2$. Thus neither part of the sharpness statement is an asymptotic or numerical assertion.

2. EXTERIOR TRIPLE LINKS

The following observation drives both cases of the proof.

Lemma 2.1. *Suppose $\delta_3(\mathcal{F}) \geq 2$, fix $E \in \mathcal{F}$, and put $X = [n] \setminus E$. For $T \in \binom{X}{3}$, define*

$$C(T) = \{e \in E : T \cup \{e\} \in \mathcal{F}\}.$$

Then $d_{\mathcal{F}}(T) = |C(T)| \geq 2$.

Proof. Every extension $T \cup \{x\}$ with $x \in X \setminus T$ is disjoint from $E \in \mathcal{F}$, so intersectingness excludes it from $\mathcal{F}$. The only possible extensions are therefore those counted by $C(T)$. $\square$

3. PROOF OF THE THEOREM

Assume for contradiction that $\delta_3(\mathcal{F}) \geq 2$. This makes $\mathcal{F}$ nonempty, so fix $E \in \mathcal{F}$ and retain the notation of Lemma 2.1.

If $n \geq 10$, then $|X| \geq 6$. Choose disjoint triples $T, T' \subseteq X$. Both $C(T)$ and $C(T')$ contain at least two points, so one may choose distinct $e \in C(T)$ and $e' \in C(T')$. The two members $T \cup \{e\}$ and $T' \cup \{e'\}$ of $\mathcal{F}$ are disjoint, a contradiction.

It remains to treat $n = 9$, so $|X| = 5$. For every $e \in E$, the triple $E \setminus \{e\}$ already has E as one extension. Choose a second extension $x_e \in X$ such that

$$(3.1) \quad (E \setminus \{e\}) \cup \{x_e\} \in \mathcal{F}.$$

If $e \in C(T)$ for $T \in \binom{X}{3}$, then

$$(3.2) \quad x_e \in T.$$

Otherwise, $T \cup \{e\}$ is disjoint from (3.1), a contradiction. Lemma 2.1 and (3.2) show that every triple of X contains at least two terms of the four-term multiset $(x_e)_{e \in E}$.

We claim that $e \mapsto x_e$ is injective. Every pair $P \subset X$, being the complement of a triple, contains at most two terms of this multiset. If some point p had multiplicity at least two and another point q occurred, then $\{p, q\}$ would contain at least three terms. If no other point occurred, $\{p, q\}$ would contain all four terms for any $q \neq p$. Both possibilities violate the pair bound, proving the claim.

Let x_0 be the unique point of $X \setminus \{x_e : e \in E\}$. Fix $j \in E$ and the triple

$$(3.3) \quad S = X \setminus \{x_0, x_j\}.$$

For any $a \in E \setminus \{j\}$, consider $Q_a = \{a, x_0, x_j\}$. An extension of Q_a by a point of X can lie in $\mathcal{F}$ only if the new point is x_a ; every other such extension is disjoint from $(E \setminus \{a\}) \cup \{x_a\}$. Thus Q_a has at most one X -extension in $\mathcal{F}$. Since $d_{\mathcal{F}}(Q_a) \geq 2$, some $b \in E \setminus \{a\}$ satisfies

$$(3.4) \quad \{a, b, x_0, x_j\} \in \mathcal{F}.$$

For every $c \in C(S)$, the edge $S \cup \{c\}$ must meet (3.4). Their X -parts are disjoint, hence $c \in \{a, b\}$. Lemma 2.1 now gives $C(S) = \{a, b\}$, and in particular $a \in C(S)$. The set S in (3.3) stays fixed while a ranges over the three elements of $E \setminus \{j\}$. It follows that $|C(S)| \geq 3$, contradicting $|C(S)| = 2$. This proves Theorem 1.1. $\square$

4. SHARPNESS

A full star is intersecting and has minimum triple degree one, so the bound in Theorem 1.1 is attained for every $n \geq 9$. The range is also best possible. On [8], let $\mathcal{B}$ consist of the following 35 four-sets (we abbreviate $\{1, 2, 3, 6\}$ as 1236):

$$\begin{aligned} \mathcal{B} = \{ & 1236, 1237, 1247, 1248, 1257, 1258, 1268, 1278, \\ & 1346, 1348, 1356, 1357, 1367, 1378, 1456, 1458, 1467, 1468, \\ & 2345, 2346, 2347, 2348, 2356, 2368, 2457, 2467, 2568, 2678, \\ & 3458, 3478, 3578, 3678, 4567, 4678, 5678\}. \end{aligned}$$

Direct exact verification gives that $\mathcal{B}$ is intersecting and that its 56 triples have degree distribution 42 of degree two and 14 of degree four. In particular, $\delta_3(\mathcal{B}) = 2$. The literal list and a definition-first verifier are included with the ancillary files.

We make no uniqueness claim for equality in Theorem 1.1; the full stars establish sharpness, not a classification of all families with $\delta_3 = 1$.

5. INDEPENDENT FINITE CERTIFICATES

For completeness, we certified the two original endpoints independently of the preceding proof. Associate a variable x_F with every $F \in \binom{[n]}{4}$. Disjoint pairs give clauses $\neg x_F \vee \neg x_{F'}$. If $x_1, \dots, x_m$ are the extensions of a fixed triple, then

$$(5.1) \quad \sum_{i=1}^m x_i \geq 2 \quad \Longleftrightarrow \quad \bigwedge_{i=1}^m \bigvee_{j \neq i} x_j.$$

Thus no auxiliary variables are needed. The exact instance sizes are:

n	variables	disjointness clauses	degree clauses
9	126	315	504
10	210	1575	840

The canonical formulas have respectively 819 and 2415 clauses. CaDiCaL 3.0.1, with seed zero, produced textual LRAT refutations. A pinned copy of the independent `lrat-check` program from the `drat-trim` project accepted both proofs. Before calling the external checker, a separate fail-closed wrapper reconstructs the canonical CNF from the definitions, validates the pinned source hashes, parses the LRAT syntax, requires a final empty-clause addition, and compiles a fresh temporary checker from the pinned source before each run. Mutation tests reject truncated, altered, non-ASCII, invalid-deletion, and cross-instance traces. Full file hashes and commands are recorded in `certificates/README.md` and the release manifest.

The $n = 8$ calculation is a positive control rather than a refutation: the same external checker rejects its SAT trace as incomplete, while the literal family above is accepted by its independent definition-level verifier.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

All instances, LRAT proofs, exact literal data, generators, independent verifiers, rejection tests, run metadata, and SHA-256 manifests accompany this article. The main theorem does not depend on a solver or on floating point computation.

OpenAI Codex assisted with literature search, proof and counterexample exploration, exact-certificate generation, adversarial checking, and drafting. Separate isolated AI roles were used as proof builder, counterexample hunter, certificate checker, and referee. No AI system is listed as an author. The named author is responsible for the statements and the released artifacts. No Lean, Coq, Isabelle, or other proof assistant was used.

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